



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO P. M. CHINNASAMI FUEL STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

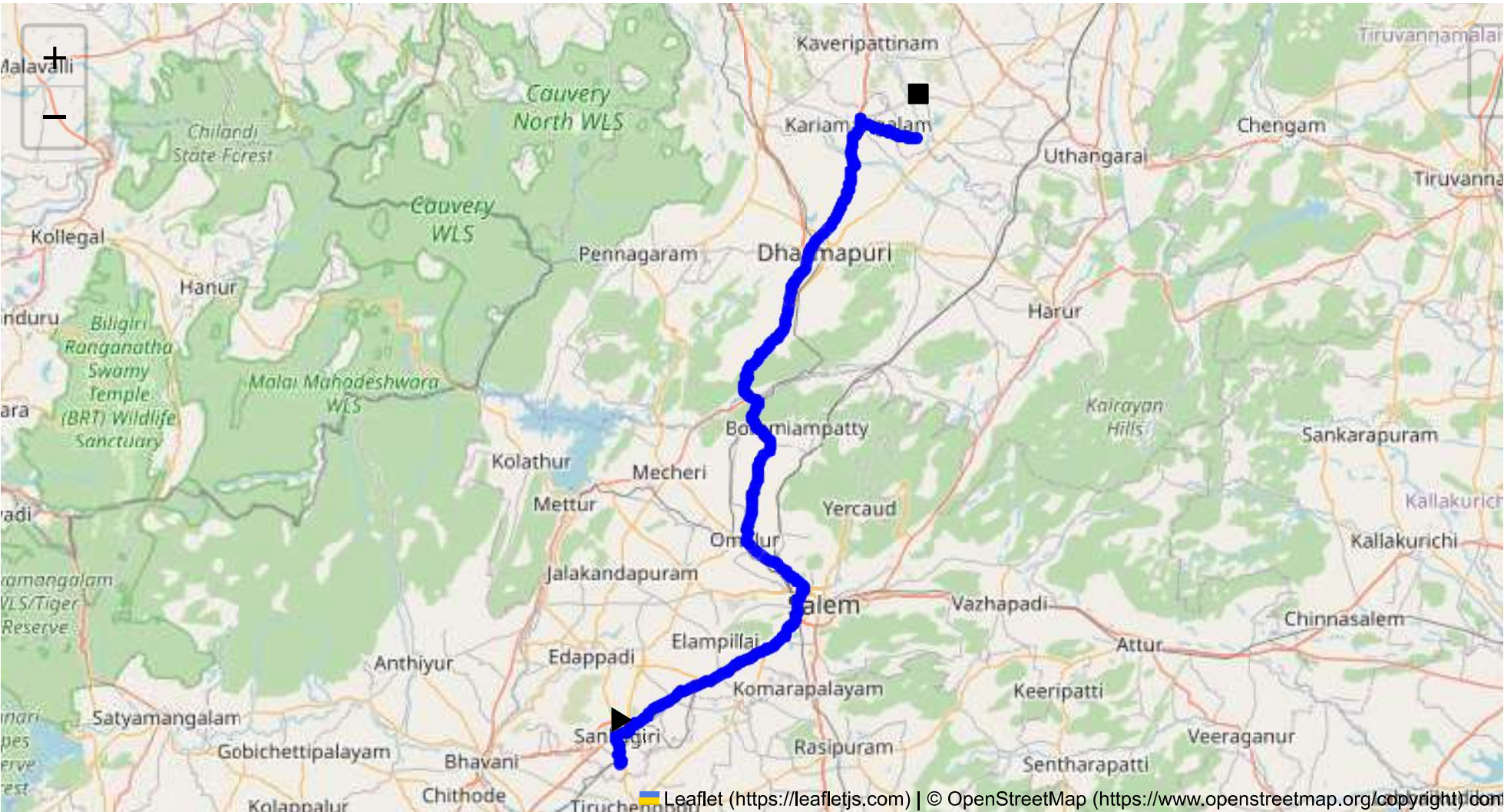
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 137.22 km
Estimated Duration: 2.6 hours
Adjusted Duration (Heavy Vehicle): 3.2 hours
Start: (11.4381, 77.8734)
End: (12.288282, 78.286353)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Chellakuttapatti covers approximately 137.22 kilometers. It primarily includes state highways and potentially national highways, with a travel time of around 2.59 hours for heavy vehicles. The journey takes you through rural and semi-urban areas, with varying road conditions.

Weather Conditions

Tamil Nadu typically experiences a tropical climate. The region can face heavy rains during the monsoon season (June to September) and extremely high temperatures in the summer months (March to May). Potential weather-related hazards include:

- **Monsoon:** Flooded roads, reduced visibility, possible landslides in hilly areas.
- **Summer:** Tire blowouts due to high temperatures, potential dehydration for drivers.

Traffic Patterns

Traffic density can vary, with peak hours generally in the morning (7 AM to 10 AM) and late afternoon/evening (5 PM to 8 PM). Major congestion areas may be around towns or intersections along the route. Regular updates from local traffic authorities can provide more specific congestion-prone areas.

Road Quality and Infrastructure

- **Highways:** Generally good quality, but look out for occasional potholes or uneven surfaces.
- **Rural areas:** May have narrow roads, limited signage, and less maintenance.
- **Bridges:** Ensure to check weight limitations, especially for heavy vehicles.

Alternative Routes

In case of emergencies, consider the following:

- **State highways:** Look for parallel state highways that might offer alternative passage.
- **Local roads:** These could be viable for detours but often require caution due to lower maintenance levels.

Local Regulations

Tamil Nadu has specific regulations regarding hazardous material transport, such as route permit requirements and time restrictions to ensure safety. It is important to comply with the designated routes and permissible hours of transit. The driver should have all necessary documentation and permits.

Historical Incidents

While data on specific historical incidents should be checked with local authorities, it is prudent to maintain vigilance in areas previously known for accidents, especially near sharp turns, intersections, or dense traffic zones.

Environmental Considerations

- **Sensitive Areas:** Watch for regions marked as forest reserves or wildlife sanctuaries. Avoid off-road driving in these areas.
- **Pollution Control:** Ensure vehicle compliance with emission norms to protect local environments.

Communication Coverage

Mobile network coverage in urban areas is generally reliable; however, rural or remote sections of the route might experience dead zones. Satellite communication or offline maps could serve as reliable backups.

Emergency Response Times

- **Urban Areas:** Swift response typically within 30 minutes to an hour.
- **Rural Areas:** Delays possible, ranging from 1 to 2 hours based on the distance to the nearest town.
- **Accident-Prone Spots:** Be cautious in areas with a history of accidents, as response might be quicker here due to pre-stationed services.

Overall Risk Assessment

The route poses moderate risks mainly due to weather conditions, variable road quality, and potential congestion at peak hours. Ensuring thorough preparation, adherence to regulations, and awareness of the local environment can mitigate many risks. Communication and responsiveness strategies should be in place to handle any emergencies efficiently.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45028, 77.87202	15 KM/Hr
4	Turn	Medium	11.45309, 77.87325	30 KM/Hr
5	Turn	Medium	11.46400, 77.87323	30 KM/Hr
6	Turn	Medium	11.46765, 77.87037	30 KM/Hr
7	Turn	High	11.47403, 77.86841	15 KM/Hr
8	Turn	High	11.49149, 77.89601	15 KM/Hr
9	Turn	High	11.49224, 77.89601	15 KM/Hr
10	Turn	Medium	11.53501, 77.95707	30 KM/Hr
11	Turn	Medium	11.53601, 77.95856	30 KM/Hr
12	Turn	Medium	11.62403, 78.10893	30 KM/Hr
13	Turn	High	11.64180, 78.11808	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
14	Turn	High	11.64325, 78.11993	15 KM/Hr
15	Turn	Medium	11.69519, 78.10775	30 KM/Hr
16	Turn	Medium	11.73958, 78.05105	30 KM/Hr
17	Turn	Medium	11.75835, 78.05016	30 KM/Hr
18	Turn	Medium	11.86847, 78.08367	30 KM/Hr
19	Turn	Medium	11.87942, 78.08197	30 KM/Hr
20	Turn	Medium	11.92585, 78.06507	30 KM/Hr
21	Turn	Medium	11.93704, 78.05402	30 KM/Hr
22	Turn	High	11.95818, 78.04978	15 KM/Hr
23	Turn	Medium	11.95988, 78.04948	30 KM/Hr
24	Turn	Medium	11.96304, 78.05136	30 KM/Hr
25	Turn	High	11.96399, 78.05135	15 KM/Hr
26	Turn	High	11.96580, 78.04923	15 KM/Hr
27	Turn	Medium	11.96675, 78.04939	30 KM/Hr
28	Turn	Medium	11.96791, 78.05070	30 KM/Hr
29	Turn	Medium	12.19881, 78.18157	30 KM/Hr
30	Turn	Medium	12.22459, 78.19136	30 KM/Hr
31	Turn	Medium	12.29110, 78.19909	30 KM/Hr
32	Blind Spot	Blind Spot	12.31460, 78.20761	10 KM/Hr
33	Turn	High	12.30795, 78.20891	15 KM/Hr
34	Turn	High	12.29976, 78.24628	15 KM/Hr
35	Turn	Medium	12.29890, 78.24676	30 KM/Hr
36	Turn	Medium	12.29768, 78.25031	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
10	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
11	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
12	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
15	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
19	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
20	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
21	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
22	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
24	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
25	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
26	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
27	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
28	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
31	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
32	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
33	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
34	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
35	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
37	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
13	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
14	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
16	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
17	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
18	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
23	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
29	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
30	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
36	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
38	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45309, 77.87325



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53501, 77.95707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53601, 77.95856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62403, 78.10893



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64180, 78.11808



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64325, 78.11993



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69519, 78.10775



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73958, 78.05105



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75835, 78.05016



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86847, 78.08367



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87942, 78.08197



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.92585, 78.06507



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93704, 78.05402



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.95818, 78.04978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95988, 78.04948



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96304, 78.05136



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96399, 78.05135



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96580, 78.04923



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96675, 78.04939



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96791, 78.05070



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.19881, 78.18157



Coordinates: 12.22459, 78.19136



Coordinates: 12.29110, 78.19909



Coordinates: 12.31460, 78.20761



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.30795, 78.20891



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.29976, 78.24628



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.29890, 78.24676



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.29768, 78.25031